

Document 2 -- Short Brief on Previous and Current Products / Tech Development

Aran Technologies | MBC-3 Registration | Max 300 words

Aran Technologies is a defence-focused UAS development team with demonstrated capability across flight controller hardware, autonomous mission software, and AI-driven surveillance systems.

Indigenous ESP32 Autonomous Flight Controller

Aran Technologies designed and fabricated a complete autonomous flight controller on the ESP32-WROOM platform, operating on a FreeRTOS dual-core architecture at 250 Hz. The system integrates a Kalman-filter IMU pipeline (MPU6050), barometer-based altitude hold (BMP280), GPS position hold and Return-to-Home (NEO-6M), and compass-referenced body-frame position control (QMC5883L). A dual-loop PID architecture -- outer angle loop, inner rate loop -- drives the motor mixer for an X-frame quadcopter. This demonstrator established indigenous design capability for flight-critical embedded systems with multi-sensor fusion.

ISR Mission Stack -- Verified Phase I Demo Build (v12-MPC-v5)

Aran Technologies developed a full autonomous ISR mission stack on PX4 SITL and Gazebo Harmonic using MAVSDK Python. Core capabilities include 360deg LiDAR obstacle avoidance driven by a Model Predictive Controller (L-BFGS-B finite-horizon QP solver), boustrophedon survey grid generation, multi-target ISR orbit sequencing, No-Fly Zone hard fencing, and a live Flask/SocketIO Ground Control Station streaming altitude, heading, battery state, and mission phase at 2.5 Hz.

The stack (v12-MPC-v5) was verified on 30 May 2026 with a full clean exit: 11-waypoint survey, primary target orbit locked at 50 m radius +/-0.5 m, RTL, and 3D occupancy map save. A 4-minute demonstration video (Gazebo and GCS side-by-side, 1920x1080) is prepared for Phase I submission. This stack forms the single-drone foundation of the MBC-3 swarm submission.

MBC-3 Collaborative Swarm -- Current Development

Five-drone hexacopter swarm operating as a distributed airborne radar, incorporating a three-layer onboard AI pipeline. Full simulation verified on ROS2 Jazzy and Gazebo Harmonic with graceful degradation, bully-protocol leader election, and real-time ASP generation.
